Spectral-Fidelity-Preserving Recalibration of Reflective Solar Bands for MERSI-II Onboard Fengyun-3D Satellite

Qiang Yu, Min Min, Di Di, Yiming Zhao, Guicai Li, Jing Li, Ling Wang, Xiuqing Hu, and Peng Zhang, *Senior Member, IEEE*

***Abstract*—Since its 2017 launch, the Medium-Resolution Spectral Imager-II (MERSI-II) aboard the China Fengyun-3D (FY-3D) polar-orbiting satellite has delivered global observations of the atmosphere, oceans, and terrestrial vegetation, yet its reflective solar bands (RSBs) exhibit gradual on-orbit radiometric degradation and temporal fluctuations. We introduce a new spectral-fidelity-preserving (SFP) cross-calibration framework that uses AQUA/moderate resolution imaging spectroradiometer (MODIS) as the reference. Unlike conventional approaches that homogenize spectral responses, the SFP model explicitly retains MERSI-II's intrinsic spectral observation characteristics while correcting radiometric bias. When applied to FY-3D/MERSI-II observations (January 2018–March 2025), the seven recalibrated RSBs achieve a mean relative bias near 1% and a 2%–3% improvement relative to the operational calibration coefficients, remaining stable throughout the 7-year record. Mann–Kendall (MK) tests identify distinct degradation breakpoints, with B05 (1.38 μm) and B06 (1.64 μm) changing earlier around October 2021, and other bands fluctuating after May 2024. We also produce daily calibration coefficients, substantially improving upon current monthly or semiannual updates. The related algorithm and daily calibration coefficients are publicly released on GitHub to support community uptake and operational use.**

***Index Terms*—Cross-calibration, Fengyun-3D (FY-3D)/Medium-Resolution Spectral Imager-II (MERSI-II), moderate resolution imaging spectroradiometer (MODIS), reflective solar bands (RSBs), spectral-fidelity-preserving (SFP) model.**

I. Introduction

HIGH-QUALITY satellite radiance observation is essential for weather forecasting and environmental monitoring. Ensuring the accuracy and long-term stability of satellite data is achieved through high-frequency radiometric calibration and continuous monitoring of onboard sensor performance [1], [2], [3], [4], [5]. Typically, most of the radiance values received by the sensors are converted into an electrical signal, which is then quantized and stored as a digital number (DN). The objective of the calibration is to ascertain the precise and stable relationship between the DN and the actual radiation value [6]. Radiometric calibration is a prerequisite for further quantitative applications affecting satellite observation data, and high-precision radiometric calibration guarantees the accuracy and usability of remote sensing data. Consequently, all satellite optical sensors undergo meticulous prelaunch calibration and testing [7], [8]. However, after launch, sensor performance, particularly calibration of visible (VIS) and near-infrared (NIR) bands (or reflective solar bands, RSBs), inevitably degrades due to mechanical stress, exposure to the harsh space environment, and other cumulative effects. To mitigate this degradation, certain satellites have been equipped with onboard calibration devices, which autonomously execute high-frequency and high-quality calibration during satellite operation. For instance, the National Aeronautics and Space Administration (NASA) polar-orbiting AQUA satellite and its moderate resolution imaging spectroradiometer (MODIS) instrument attained VIS and NIR band radiation calibration accuracies within an average of 3%, while the MODIS VIS and NIR calibration accuracies aboard the TERRA satellite were 5% and 3% [9], [10], [11], [12], respectively. Its successor, the latest visible infrared imaging radiometer suite (VIIRS) onboard Joint Polar Satellite System satellite (JPSS), has further improved radiometric uncertainty, maintaining VIS and NIR calibration within higher 1.5% [13], [14], [15], [16]. Calibration practices of these high-performance sensors demonstrate that high-frequency calibration is critical for maintaining long-term data accuracy, and existing technologies already enable

Received November 8, 2025; revised January 22, 2026; accepted February 28, 2026. Date of publication 6 March 2026; date of current version 16 March 2026. This work was supported in part by the National Key Research and Development Program of China under Grant 2023YFF0805401, in part by the Natural Science Foundation of China under Grant 42175086, in part by the Innovation Group Project of Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai) under Grant SML2023SP208, and in part by the Science and Technology Planning Project of Guangdong Province under Grant 2023B1212060019. *(Corresponding author: Min Min.)*

Qiang Yu and Min Min are with the School of Atmospheric Sciences, Guangdong Province Key Laboratory for Climate Change and Natural Disaster Studies, and Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai), Sun Yat-sen University, Zhuhai 519082, China (e-mail: minm5@mail.sysu.edu.cn).

Di Di is with the Key Laboratory for Aerosol-Cloud-Precipitation of China Meteorological Administration, School of Atmospheric Physics, Nanjing University of Information Science and Technology, Nanjing 210044, China.

Yiming Zhao is with the State Key Laboratory of Environment Characteristics and Effects for Near-Space, Beijing Institute of Technology, Beijing 100081, China.

Guicai Li, Ling Wang, and Xiuqing Hu are with the Key Laboratory of Radiometric Calibration and Validation for Environmental Satellites (LRCVES/CMA), National Satellite Meteorological Center, China Meteorological Administration (NSMC/CMA), Beijing 100081, China.

Jing Li is with the Department of Atmospheric and Oceanic Sciences, School of Physics, Peking University, Beijing 100871, China.

Peng Zhang is with China Meteorological Administration (CMA), Meteorological Observation Center, Beijing 100081, China.

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2026.3671371, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2026.3671371

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:17:06 UTC from IEEE Xplore. Restrictions apply.

daily or even higher-frequency calibration updates, providing a reference direction for calibration optimization of other sensors.

Besides, traditional on-orbit radiometric calibration strategies for Earth observation satellites encompass a variety of alternative approaches, including the use of pseudo-invariant ground targets for vicarious calibration (VC), lunar observations to exploit the Moon's exceptional radiometric stability, and simultaneous nadir overpass (SNO) techniques that enable cross calibration by matching near-simultaneous observations from different platforms. In recent years, the spectral band adjustment factor (SBAF) methodology, supported by some high-precision radiative transfer models (RTMs), has become a key tool to reconcile differences in spectral response functions (SRFs) between instruments [17]. This enables more accurate transfer of calibration coefficients between sensors with mismatched bands, particularly in cross-sensor coordination tasks. For polar-orbiting satellites such as China's FY-3 series satellites, the National Satellite Meteorological Center (NSMC) of the China Meteorological Administration (CMA) routinely performs VC using multiple stable reference targets. These include well-characterized ground-based sites (e.g., China Dunhuang Gobi calibration field) [18], [19], [20] as well as invariant targets such as deep convective clouds (DCCs) [21]. This comprehensive on-orbit calibration approach significantly enhances the quantitative application potential of Fengyun meteorological satellite data products. Meanwhile, the Global Space-based Inter-Calibration System (GSICS), an important component of the Global Earth Observation System of Systems (GEOSSs), has institutionalized cross-calibration practices by generating reference datasets that link target sensor observations to well-characterized instruments such as MODIS and VIIRS, enabling systematic correction of long-term drifts and intersensor biases [12], [15].

Since the successful launch of China's new-generation FY-3A in 2008, the FY-3 series of polar-orbiting meteorological satellites has carried the moderate resolution imaging spectroradiometer (MERSI), initially featuring 20 bands and later upgraded to the 25-band Medium-Resolution Spectral Imager-II (MERSI-II) sensor beginning with the FY-3D satellite in November 2017. Unlike well-equipped instruments such as MODIS or VIIRS, the MERSI-II lacks a dedicated onboard radiometric calibration system due to inherent constraints in satellite design and sensor architecture. As a result, alternative calibration approaches, such as field campaigns and monitoring of pseudo-invariant calibration targets (e.g., deserts like the China Dunhuang Gobi site), have been employed [18], [22], [23], [24]. However, these alternative calibration methods introduce considerable and unpredictable uncertainties in their implementation and are typically performed at low temporal frequencies (6–12 month intervals), which fails to capture short-term radiometric fluctuations of sensors (e.g., sudden response changes caused by optical component contamination) and cannot meet the needs of applications requiring high-frequency and high-precision data, such as cloud detection and vegetation dynamic monitoring.

To address the aforementioned issues and challenges, as the first satellite carrying MERSI, FY-3A serves as a radiometric reference for subsequent Fengyun-3 series satellites. Previous studies [5], [25], [26] have predominantly employed the VC approach, which integrates multiple targets to track variations in instrument performance. During calibration in RSBs, the accuracy was validated against AQUA/MODIS and remained within 3% [27]. The RSBs for FY-3B employ a multisite calibration scheme to achieve a wide dynamic VC range on orbit. This scheme leverages the existing FY-3A calibration result to ensure consistent transfer and verification. After the recalibration process, the RSBs obtained for FY-3B maintain calibration accuracy within about 3% [22], [27]. The RSBs for FY-3C use a multicross calibration method that includes an SNO with MODIS and tracking of pseudo-invariance targets. During the consistency transfer calibration process, FY-3A first serves as the reference for the calibration of FY-3B. Then, FY-3C/MERSI is recalibrated using the calibrated FY-3B as the reference. With the exception of a downward drift observed in some bands, the calibration accuracy of the FY-3C RSBs remains within the same precision range as the FY-3A/B RSBs [5], [25]. This status quo highlights the urgency of developing a high-frequency calibration scheme (especially daily updates) for MERSI-II; only daily calibration can timely capture short-term radiometric fluctuations and fill the gap of existing methods in temporal resolution.

This study focuses on the 25-band MERSI-II instrument onboard the FY-3D satellite, as recalibration efforts for its predecessors (FY-3A/B/C) have already been completed [27]. As noted previously, the absence of onboard calibration systems [24], [28] fundamentally constrains the calibration accuracy and long-term stability of FY-3D/MERSI-II. To mitigate these limitations, high-performance reference sensors, such as MODIS (a sensor with well-validated radiometric stability), are commonly employed for onboard cross calibration to ensure radiometric consistency among different instruments [29], [30], [31]. However, traditional SNO-based alternative calibration approaches tend to transfer the spectral response characteristics of the reference instrument (e.g., MODIS) to the target sensor, leading to spectral homogenization and thereby compromising the preservation of the target sensor's unique radiometric properties [32], [33], [34]. To overcome this issue, we propose a novel spectral-fidelity-preserving (SFP) cross-calibration method in this investigation. This approach enables high-frequency calibration in the RSBs of FY-3D/MERSI-II while simultaneously inheriting the radiometric accuracy of the reference sensor and preserving the inherent spectral fidelity of the target instrument. To resolve the issue of inadequate calibration frequency, the model's 15-day calibration results are further coupled with suitable interpolation algorithms, thereby facilitating the acquisition of daily calibration results. Therefore, the primary objective of this study is to establish a consistent dataset of calibration coefficients for the FY-3D/MERSI-II RSBs, featuring long-term sequences and high frequencies. Moreover, this methodology is also applicable for cross calibration among various satellite sensors.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:17:06 UTC from IEEE Xplore. Restrictions apply.

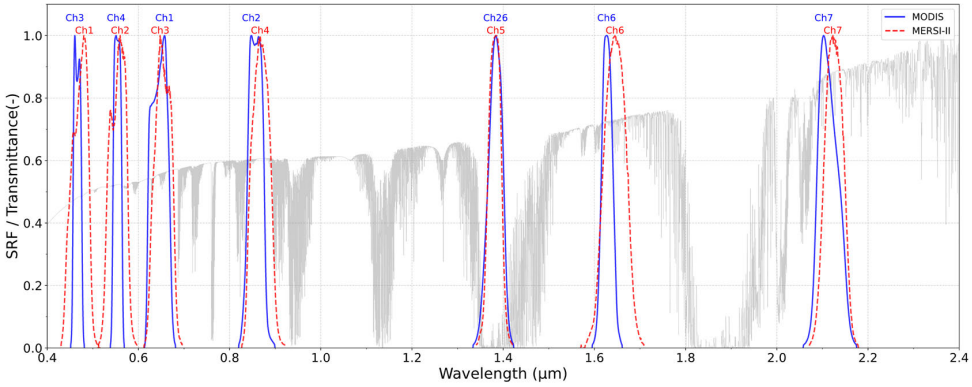


Fig. 1. SRFs of seven RSBs of FY-3D/MERSI-II (red dashed line) and AQUA/MODIS (blue solid line). The gray line is the simulated atmospheric transmittance by using the MODerate resolution atmospheric TRANsmission (MODTRAN) and 1976 U.S. standard atmospheric profile.

The remainder of this article is organized as follows. Section II provides a brief description of the instrumentation and dataset. Section III offers a detailed description of the full cross-calibration workflow of the combined SFP model. Section IV presents the final recalibration results, and Section V discusses the implications, comparisons, and future directions. Section VI summarizes the conclusions and new findings of this study.

II. Data and Model

This study uses radiance observations from the second-generation MERSI-II aboard FY-3D, launched on November 15, 2017, into a sun-synchronous orbit with an afternoon ascending node near 13:30 local time. Compared with its predecessor, MERSI-II expands spectral coverage from 20 to 25 bands. In this study, we primarily use the FY-3D/MERSI-II Level-1B (L1B) radiance product (1KM) and the companion geolocation product (1KM_GEO) from January 2018 to March 2025. The L1B datasets were produced from raw Level-1A (L1A) measurements via geometric correction, navigation processing, and radiometric calibration. Seven FY-3D/MERSI-II solar reflective bands were recalibrated, with central wavelengths at 0.47, 0.55, 0.65, 0.86, 1.38, 1.64, and 2.13 μm (Bands 1–7, respectively).

The primary reference sensor used in this study is the MODIS aboard the AQUA satellite. Launched on May 4, 2002, the AQUA satellite follows a sun-synchronous orbit with an equatorial crossing time of approximately 13:30 local time. MODIS features 36 spectral bands covering a broad wavelength range from visible (0.4 μm) to thermal infrared (14.4 μm), making it highly suitable for cross calibration with other sensors. Furthermore, a close examination of the orbital parameters of the FY-3D and AQUA reveals a notable similarity, with the local time at ascending node (LTAN) occurring at 13:30 and orbital inclinations of 98.75° and 98.2°, respectively. Given the similar orbital configurations of AQUA and FY-3D, a sufficient number of collocated observations between MODIS and MERSI-II can be obtained. Additionally, the spectral band arrangements and response characteristics of MODIS exhibit strong similarities to those of MERSI-II, simplifying spectral band matching between them. Fig. 1 illustrates the SRFs of the seven corresponding RSBs for both instruments. Because of these factors, MODIS serves as an appropriate reference for cross calibration, effectively reducing the complexity typically associated with radiometric calibration procedures. Besides, MODIS, as the reference sensor, is also highly recommended by the GSICS community for spaceborne cross calibration [35].

Since the MODTRAN model (Version 4.2) operates on the correlated-k approximation and employs layer-by-layer radiative integration, this advanced RTM requires comprehensive atmospheric profile data to ensure the accuracy and completeness of the final simulation results [17], [36], [37]. It covers a wide spectral range from 0.2 to 100 μm with a fine resolution of 1 cm^{-1}, and comprehensively accounts for key radiative transfer processes, including molecular absorption, aerosol and cloud scattering, and surface reflection. During the RTM and other multispectral imager calibration coefficients, the operation is typically performed using the 83 independent atmospheric profiles provided by the European Center for Medium-Range Weather Forecasts (ECMWF) in 2008 (the download URL for these atmospheric profiles is: https://nwp-saf.eumetsat.int/downloads/profiles/ECMWF_83P_91L.dat). These atmospheric profiles have undergone an expansion to encompass six representative types: tropical, mid-latitude summer, mid-latitude winter, subpolar summer, subpolar winter, and the 1976 US standard atmospheric profile. This augmentation facilitates adaptation to more complex application scenarios. Each profile includes vertically resolved atmospheric state variables, such as temperature, pressure, water vapor concentration, and ozone distribution, spanning from the Earth's surface to the upper atmosphere. Concurrently, to align with the computational demands of RTM, the envelope was segmented into 101 layers, categorized based on pressure zones, with atmospheric pressure ranging from 0.005 to 1100 hPa at constant pressure intervals. This configuration furnishes the foundational support for subsequent precise simulations.

III. Methodology

This study primarily consisted of two main tasks: first, the construction of the SFP model, and second, the integration of long-term and high-frequency recalibration using a reference sensor with the model. Fig. 2 illustrates the flowchart of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:17:06 UTC from IEEE Xplore. Restrictions apply.

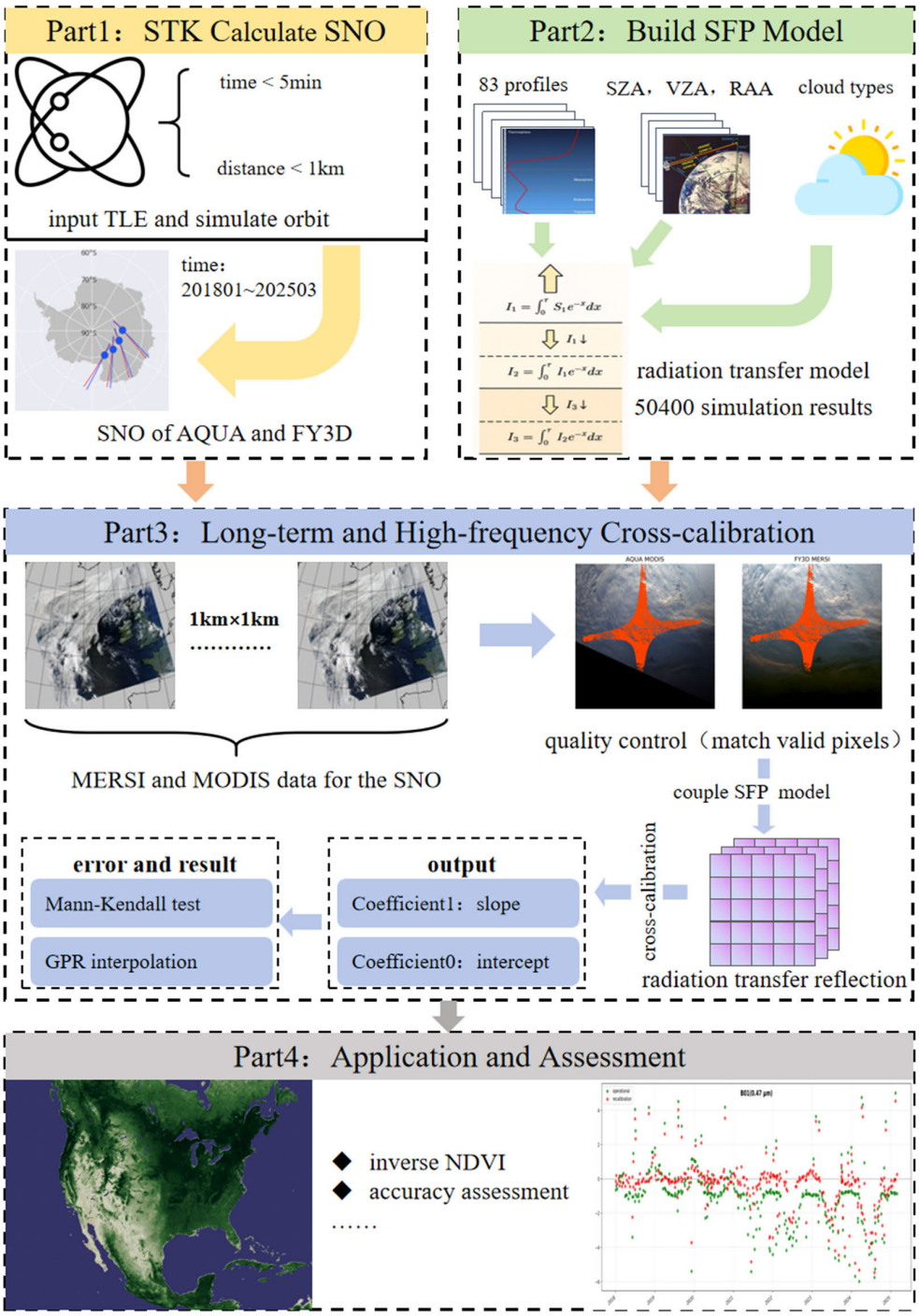


Fig. 2. Flowchart schematics of the SFP-based recalibration method.

the proposed SFP-based recalibration method, which includes the following key steps: establishment of the SFP model, orbit crossover computation, cross calibration of RSBs, quality assessment of calibration performance, and validation of Level-2 science products.

A. SFP Model

In this study, we used classical MODTRAN as the core forward RTM to simulate top-of-atmosphere (TOA) radiance observed by a satellite sensor, chosen for its extensive validation, rigorous physical basis, and high spectral fidelity. To ensure diversity and global representativeness, we assembled 83 atmospheric profiles from the ECMWF archive spanning multiple climatic zones; these profiles were then filtered and matched to surface types to minimize computational redundancy while maximizing surface-condition variability, thereby improving the efficiency and coverage of the simulations. In consideration of the characteristics inherent to RSBs, the construction of the model necessitates the coverage of the spectral range from 0.3 to 3 μm, with a resolution of 1 cm^{-1}.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:17:06 UTC from IEEE Xplore. Restrictions apply.

The core physics of RTM is based on the monochromatic Radiative Transfer Equation (RTE), which describes the evolution of radiance along an atmospheric path

$$\mu \frac{\mathrm{dL}_\lambda(\tau,\mu,\phi)}{\mathrm{d}\tau} = \mathrm{L}_\lambda(\tau,\mu,\phi) - \frac{\sigma(\lambda)}{\sigma(\lambda)+\alpha(\lambda)} \frac{1}{4\pi} \int_0^{4\pi} \mathrm{P}_\lambda\left(\Omega',\Omega\right) \mathrm{I}_\lambda\left(\Omega',\Omega\right) \mathrm{d}\Omega \quad (1)$$

where $\mathrm{L}_\lambda(\tau,\mu,\phi)$ is the radiance at the corresponding optical depth τ and wavelength λ in the direction (μ,ϕ). μ is the cosine of the view zenith angle. $(\sigma(\lambda))/(\sigma(\lambda)+\alpha(\lambda))$ is the single scattering albedo; σ is the scattering coefficient and α is the absorption coefficient, and $P_\lambda(\Omega',\Omega)$ is the phase function, describing scattering from direction Ω' (incident direction) to Ω (emitted direction).

The following discussion examines atmospheric transmittance (T_λ) and optical thickness (τ_λ) in RTE, as these quantities strongly govern the radiance measured by spaceborne sensors, particularly in the RSBs. These two variables can be expressed as follows:

$$T_\lambda = e^{-\tau_\lambda} \quad (2)$$

$$\tau_\lambda = \int_{s_0}^{s} \kappa_\lambda\left(s'\right) ds' \quad (3)$$

where T_λ is the atmospheric transmittance at the wavelength λ. The exponential term $-\tau_\lambda$ follows the Beer–Lambert law, which describes attenuation effect of radiation through an absorbing and scattering medium. The τ_λ is the optical depth at the wavelength λ, which quantifies the cumulative attenuation effect of the atmosphere along the path from s_0 to s. The $\kappa_\lambda(s')$ is extinction coefficient per unit length, accounting for absorption and scattering processes. The aerosols and clouds employed in model construction use predefined physical parameters. The corresponding τ values are as follows: ocean aerosol = 0.17, desert aerosol = 0.78, cumulus = 216, stratus = 38, altostratus = 76, nimbostratus = 46. It is important to note that the optical thickness of aerosols is representative of the conditions specified in this study alone, namely, a wind speed of 6.9 m/s and visibility of 23 km. The custom standard cirrus was selected as optical depth of 1.5, representing an optically thin cloud case.

According to these formulas, the radiance measured by RSBs of spaceborne sensors can be calculated as follows:

$$L_{sensor} = L_{surface} \times T_{atm} + L_{path} \quad (4)$$

where L_{surface} is the band-averaged radiance reflected or emitted by the surface; T_{atm} denotes the band-averaged atmospheric transmittance; L_{path} is the band-averaged path radiance induced by solar light scattering, including Rayleigh scattering (molecular scattering), aerosol scattering, and cloud scattering. As the core RTM, MODTRAN explicitly accounts for Rayleigh scattering in the short-wave bands (e.g., VIS and NIR), so L_{path} inherently incorporates this nonnegligible component for RSBs.

Under realistic atmospheric radiative transfer conditions, satellite observation geometry is a primary determinant of observed TOA radiance. Rather than enumerating all scattering pathways, we characterize the illumination–viewing configuration by the solar zenith angle (SZA) and solar azimuth angle (SAA), and the sensor view zenith angle (VZA) and view azimuth angle (VAA). These parameters determine atmospheric path length and the bidirectional distribution of incident and reflected radiation. Large VZA values (common at high latitudes and near swath edges) increase perspective distortion, reduce the surface contribution to the measured signal, and complicate reflectance calibration [38]. The relative azimuth angle (RAA = |SAA − VAA|) further modulates anisotropy through bidirectional reflectance effects, producing pronounced variations in observed TOA radiance. It is important to acknowledge that SAA is set to 0 by default; however, RAA is the determining factor in the simulation's accuracy. The RAA has been demonstrated to cover most observation scenarios within the 0°–180° range, thereby ensuring the reliability and efficiency of setting observation angles in simulations.

Because atmospheric impacts are strongly wavelength dependent, especially in the VIS and NIR bands, the simulations include representative aerosol models (e.g., marine and dust) with distinct extinction and scattering characteristics, as well as common cloud types (e.g., cirrus, altostratus, cumulus, stratocumulus, and cumulonimbus) [39]. Table I summarizes the MODTRAN input variables used to construct the SFP model: angular geometry (SZA, VZA, RAA), aerosol and cloud models, surface conditions, and spectral settings. Combining six representative cloud types, two aerosol models, 45 illumination–viewing geometries, and 83 atmospheric profiles yields a total of 50 400 simulation cases. This dataset is designed to provide realistic, reliable, and accurate support for developing and validating the SFP model.

It is important to emphasize that MODTRAN does not fully capture the dynamic variations in surface reflectance. The primary objective of the MODTRAN-based simulations in this study is not to replicate real-world surface conditions, but rather to quantify the relative radiometric differences between MODIS and MERSI-II across various surface types under controlled atmospheric and geometric conditions. In this context, MODTRAN, as a physics-based tool, can isolate sensor-related spectral effects while minimizing confounding factors. By performing paired simulations using identical high-spectral-resolution input data, mutual calibration coefficients can be derived, ensuring that the final differences are entirely driven by sensor SRF characteristics rather than surface variations.

To further assess the robustness of this approach, an independent validation was conducted using real hyperspectral surface reflectance data from the USGS spectral library. These spectra were convolved with the SRFs of both sensors to directly evaluate relative spectral differences without relying on atmospheric simulations. The comparison demonstrates that the relative radiometric discrepancies obtained from hyperspectral convolution are consistent with those derived from MODTRAN-based simulations (see Supplementary Material Fig. S1), supporting the adequacy of the proposed method for SRF-related intercalibration.

While coupling MODTRAN with a full land surface RTM (e.g., SAIL or DART) could further enhance surface realism, such complexity is not essential for the specific purpose of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:17:06 UTC from IEEE Xplore. Restrictions apply.

TABLE I
INPUT VARIABLES OF THE MODTRAN MODEL FOR CONSTRUCTING THE SFP MODEL

Parameter	Settings
Surface types	snow, forest, farm, desert, urban, ocean, grass, tundra
Aerosol types	ocean, desert
Cloud types	clear-sky, cumulus, stratus, altostratus, nimbostratus, cirrus
SZA	0°, 30°, 45°, 55°, 60°
SAA	0°
VZA	0°, 30°, 45°, 55°, 60°
VAA	0°, 45°, 90°, 135°, 180°

Note: Solar zenith angle = SZA, Solar azimuth angle = SAA, sensor view zenith angle = VZA and sensor view azimuth angle =VAA.

SRF mismatch correction addressed here. The supplementary hyperspectral analysis provides an empirical benchmark, confirming that the MODTRAN-based approach captures the dominant spectral effects relevant to cross-sensor recalibration.

MODTRAN-simulated hyperspectral TOA radiances provide physically consistent References for bridging MODIS and MERSI-II observations, enabling accurate atmospheric correction and spectral harmonization. This approach enhances spectral fidelity within the cross-calibration workflow. To transfer radiometric calibration from a reference sensor, the high-spectral-resolution simulations are projected onto each instrument's bandpass by convolving them with its RSBs SRFs. The band-averaged effective radiance measured by a sensor in the RSBs is given by

$$L = \frac{\int_{\lambda_2}^{\lambda_1} L(\lambda) * R(\lambda)\, d\lambda}{\int_{\lambda_2}^{\lambda_1} R(\lambda)\, d\lambda} \tag{5}$$

where $L(\lambda)$ is the high-spectral-resolution TOA radiance simulated with MODTRAN, and $R(\lambda)$ denotes the normalized SRFs of the sensor's RSBs. This convolution enforces spectral consistency between the simulated and observed radiances by incorporating the sensor-specific bandpass characteristics.

Building on the hyperspectral simulations (from 0.3 to 3 μm with a spectral resolution of 1 cm^{-1}) and SRF convolutions described above, we matched MODIS and MERSI-II band-equivalent radiances to derive an SFP model for each RSB. The transmittance module evaluated linear and polynomial regressions and selected the final form by comparing correlation coefficients; the linear model performed best overall, although MERSI-II B06 (1.64 μm) exhibited noticeably weaker fits in Fig. 3, consistent with strong water-vapor absorption in its passband [40], [41]. Fig. 3 shows the resulting linear SFP fits for the seven RSBs using Aqua/MODIS-equivalent radiances. These fits were then used to construct intersensor transfer functions that quantify the radiometric relationship between MODIS and MERSI-II, providing the basis for subsequent cross calibration and reflectance harmonization.

With regard to the process of converting radiation values from reference sensors to target sensors, it should be noted that, since spectral deviation correction has already been performed, only the fitting operation is described here. The unified spectral fitting formula in the SFP model is as follows:

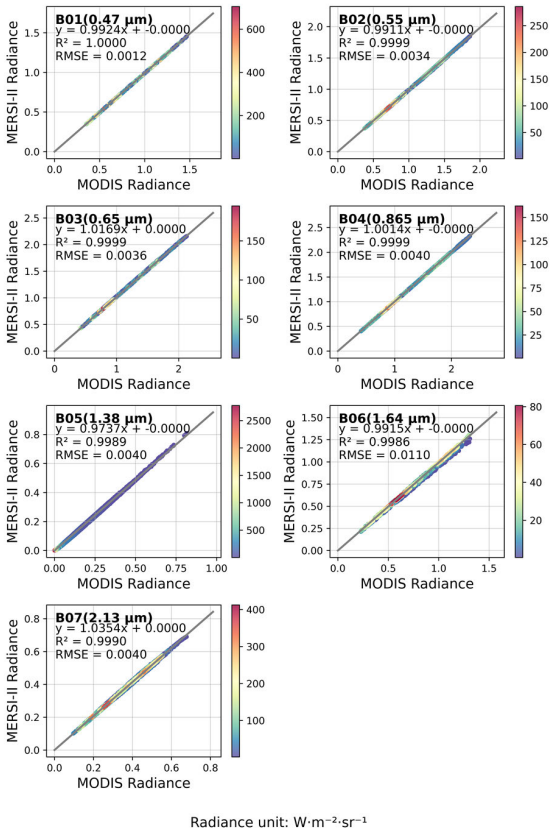


Fig. 3. Linearly fit SFP models for the seven RSBs of FY-3D/MERSI-II, derived from the equivalent simulated radiances measured by AQUA/MODIS.

$$y(\lambda) = \sum_{i=0}^{m} a_i \phi_i(\lambda) \tag{6}$$

where $y(\lambda)$ is the band-averaged radiance value of target sensor. The $\phi_i(\lambda)$ is the basis function, represents the band-averaged radiance value of reference sensor, for polynomial fitting use $\phi_i(\lambda)$ higher-order power. The a_i coefficients to be estimated; m is polynomial order.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:17:06 UTC from IEEE Xplore. Restrictions apply.

B. Long-Term and High-Frequency Cross Calibration

As summarized in Fig. 2, after completing the radiative transfer simulations, we implemented the core cross-calibration procedure. The workflow has three parts: 1) computation and selection of SNOs; 2) the operational steps of the cross calibration; and 3) derivation and screening of calibration coefficients to meet the specified requirements. Fig. 2 outlines the process; the remainder of this section elaborates on each component.

Cross calibration is feasible provided that sufficient SNOs are available, as FY-3D/MERSI-II and AQUA/MODIS occupy similar sun-synchronous orbits (inclination ≈ 98°; local time of ascending node ≈ 13:30), enabling frequent cross-view scenes. We use their two-line element sets (TLEs) from January 2018 to March 2025 to calculate SNO events. Candidate SNOs were defined by an intersensor acquisition time difference ≤ 5 min and a ground-track separation ≤ 5 km [6], [30]. These temporal and spatial thresholds represent the default values within the GSICS system, thereby ensuring that observation conditions for the two satellites do not undergo significant abrupt changes. After screening, valid SNOs were retained, and the corresponding MERSI-II and MODIS L1B granules were batch-downloaded by timestamp.

As described above, SNO observations were acquired semi-monthly at the beginning and near the middle of each month (~15-day interval), and used as training data; test samples were taken from the midpoint between the two acquisition dates. After preliminary screening (scene uniformity checks, solar-flare filtering, and masking of cloud-edge pixels), the data underwent the detailed quality-control procedure shown in Fig. 2. Cross calibration was then performed on valid pixels and required two standard corrections: 1) dark-current removal by subtracting deep-space counts and 2) heliocentric-distance normalization to 1 Astronomical Unit (AU). The final corrected $DN_{correct}$ value, accounting for both effects, is given by

$$DN_{correct} = (DN - DN_{dark}) \times d^2 \tag{7}$$

where DN is the original DN, DN_{dark} is the deep space view count value (dark current), and d is the Sun–Earth distance in AU.

These corrections ensure that the calibration inputs represent physical radiance uncontaminated by instrumental offsets and Sun–Earth distance effects. After quality control and bias removal, the data were ingested by the calibration module. To preserve spectral fidelity, we used the new SFP model mentioned in Section III-A to map radiance between targeted and reference instruments: the estimated transfer coefficients were applied to MODIS matchups to generate MERSI-II–equivalent radiance, which was then converted to reflectance for calibration. The core calibration relation is written as follows:

$$\rho = \frac{\pi * (a * L_{modis} + b)}{E_0} \tag{8}$$

where L_{modis} denotes the radiance measured by MODIS. Both a and b coefficients are derived from the SFP model (Fig. 3), and E_0 is the corresponding band-averaged solar constant of RSBs of FY-3D/MERSI-II.

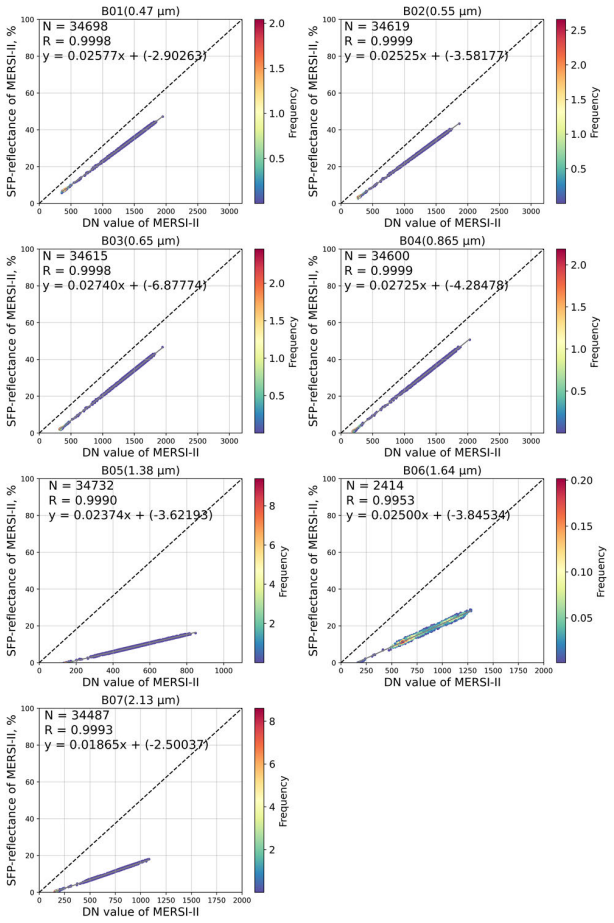


Fig. 4. Daily cross-calibration results (y = slope*(x) + intercept) of seven RSBs of FY-3D/MERSI-II on October 3, 2018 based on the reference sensor of AQUA/MODIS and the SFP model. N and R, respectively, represent the total sample number and correlation coefficient. In particular, Band 06 (1.64 μm), which is strongly affected by atmospheric water vapor absorption, undergoes stricter filtering to avoid unstable radiometric behavior, resulting in a reduced number of valid samples.

The foregoing procedure produced preliminary reflectance-calibration results for 253 SNO cases from January 2018 to March 2025. These were partitioned into a test set and a validation set; the validation set was used to screen outliers and remove unreliable test cases, yielding the final high-frequency recalibration series. Fig. 4 illustrates a valid daily calibration example on October 3, 2018, showing the recalibration results for Bands 1–7 of FY-3D/MERSI-II. Because the applied quality control criteria (e.g., cloud-edge masking, spectral consistency checks, and atmospheric sensitivity screening) act differently across spectral bands, the number of valid SNO samples is not uniform among bands. Note that 4096 is the theoretical maximum measurement value for MERSI-II. In practice, however, the instrument saturates at approximately 3500. Therefore, during the SNO process, the DN value should be kept between 0 and 3500. Within this range, the calibration coefficient demonstrates excellent adaptability. This effect is particularly pronounced for shortwave infrared bands located in water vapor absorption regions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:17:06 UTC from IEEE Xplore. Restrictions apply.

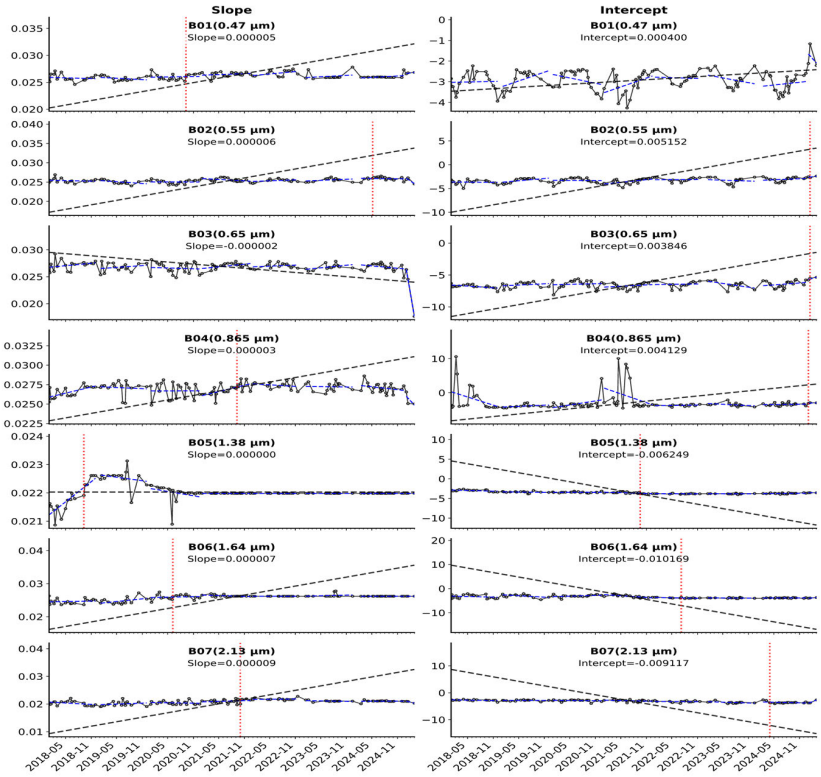


Fig. 5. MK test results for slopes and intercepts in the calibration outcomes of Bands 1–7 of FY-3D/MERSI-II, where the red, blue, and gray dashed lines, respectively, indicate the time point of mutation occurrence, decay rate of the year, and median trendline.

IV. Result

A. Statistical Analysis of Long-Term Recalibration Results

Following the procedures described above, we derived recalibration coefficients for the VIS and NIR bands of FY-3D/MERSI-II at a 15-day interval from January 2018 to March 2025. We then statistically characterized the time series of calibration parameters—slope (gain) and intercept (offset)—for each spectral band and assessed their consistency with previously reported results [27].

To quantify monotonic trends in the recalibration time series, we employed the nonparametric Mann–Kendall (MK) test [42], chosen for its robustness to nonnormality, outliers, and missing values. The MK procedure computes the Kendall score S and its variance $Var(S)$, then forms a standardized statistic Z to assess significance. Its fundamental formula is written as follows:

$$Z = \begin{cases} \dfrac{S-1}{\sqrt{Var(s)}}, & if\ S > 0 \\ 0, & if\ S = 0 \\ \dfrac{S+1}{\sqrt{Var(S)}}, & if\ S < 0 \end{cases} \tag{9}$$

where S is the MK statistic computed from all pairwise comparisons in the time series. Z is the standardized normal deviate [approximately N(0,1)], which is used to assess trend significance.

As shown in Fig. 5, the calibration coefficients of the seven MERSI-II RSBs exhibit statistically significant temporal trends from January 2018 to March 2025 based on the MK test. For each band, the gray dashed line indicates the median trend estimated by Sen's slope. Bands B01 (0.47 μm), B02 (0.55 μm), and B04 (0.86 μm) show clear positive trends, whereas B03 (0.65 μm) shows a negative trend. Notably, B05 (1.38 μm) shows essentially no trend, suggesting no appreciable performance change. All reported trends are significant at the 95% confidence level.

The recalibration slopes of B05 (1.38 μm) and B06 (1.64 μm) exhibited earlier mutations around October 2021, with both bands situated in proximity to the water vapor absorption band. All sudden changes of other bands occurred in May 2024 or later, despite the planned operational lifespan of the FY-3D satellite being only 5 years.

Additionally, Fig. 5 illustrates the annual attenuation trends. As can be seen, all bands except B03 (0.65 μm) showed significant attenuation during the initial phase. Prior to 2022, calculations indicate that the yearly degradation rates for the six channels excluding B03 (0.65 μm) generally ranged from 0.0035% to 0.005%. Since then, the rates have largely remained below 0.003%. This attenuation pattern more accurately reflects the radiation decay and performance variations of the MERSI-II instrument across its operational phases.

B. Optimal Interpolation Method for Long-Term Recalibration Coefficients

For quantitative applications of MERSI-II data, high-frequency calibration coefficients are required for the RSBs of FY-3D/MERSI-II along each swath. Because these coefficients exhibit temporal variability and nonlinear behavior, careful selection of the interpolation scheme is essential for a 15-day interval recalibration scheme. Using an independent

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:17:06 UTC from IEEE Xplore. Restrictions apply.

validation dataset, we compared linear interpolation, cubic splines, and Gaussian process regression (GPR) [43]. Quantitative results in Fig. 6 show that the root-mean-square error (RMSE) of GPR interpolation is 0.0012–0.0025 (slope) and 0.0008–0.0018 (intercept), which is more than 40% lower than that of linear interpolation (RMSE: 0.0038–0.0052) and cubic spline interpolation (RMSE: 0.0045–0.0061). Moreover, the deviation between the interpolated coefficients and the 15-day interval SNO-measured coefficients is $\leq 0.8\%$.

This study adopted GPR as the method for high-frequency interpolation of recalibration coefficients. Currently, the relevant interpolation results, detailed usage instructions, and GPR-based interpolation code have been packaged and uploaded to GitHub. The recalibration coefficients (including daily-generated results) for the seven RSBs of FY-3D/MERSI-II are publicly available at https://github.com/Rhett12345/mersi_recalibration, facilitating direct application of high-precision daily calibration data for global users.

The fundamental formulation of GPR rests on the Gaussian-process assumption: the joint distribution of all observed and predicted points is multivariate normal. Accordingly, the conditional (posterior) distribution of the predictions given the observations is also Gaussian. In the expression below, "|" denotes conditioning and "~" denotes "is distributed as"

$$f_*|X, y, X_* \sim N(\mu_*, \Sigma_*) \tag{10}$$

where f_* is the function value at the prediction point, and $N(\mu_*, \Sigma_*)$ is a multivariate Gaussian distribution. The prediction output is generated given the training input X, the observed value y, and the test point location X_*.

The predicative distribution of GPR with mean and covariance is given as follows:

$$\mu_* = K(X_*, X)\left[K(X, X) + \sigma_n^2 I\right]^{-1} y \tag{11}$$

$$U = K(X_*, X_*) - K(X_*, X)\left[K(X, X) + \sigma_n^2 I\right]^{-1} K(X, X_*) \tag{12}$$

where $K(X, X)$ is the covariance matrix of the training points, $K(X, X_*)$ is the cross-covariance between training and test points, $K(X_*, X_*)$ is the covariance among test points, and $\sigma_n^2 I$ accounts for the observation noise. μ_* represents the predicted values at the test points, and U reflects the predictive uncertainty, indicating the confidence of the model in its predictions. The superscript "$*$" indicates that these values are all predictions, while the subscript "n" denotes noise.

The interpolated recalibration coefficients from January 2018 to March 2025 are displayed in Fig. 7 (slope) and Fig. 8 (intercept). A thorough analysis of the interpolation results reveals that the majority of test points and validation points are contained within the 95% confidence interval. Furthermore, to enhance the reliability of the interpolation results, the start and end dates of the calibration results were each extended by 15 days. Pseudo-samples were generated for the periods from December 15, 2017, to December 31, 2017, and from March 1, 2025, to March 15, 2025. This approach effectively achieves data augmentation.

C. Assessments of Recalibration Errors

Following the optimal interpolation process in Section IV-B, it is imperative to undertake a more comprehensive quantification of the recalibration accuracy. In this study, MODIS reflectance was used as the baseline reference value. The MERSI-II reflectance was calculated using two sets of recalibration coefficients: operational calibration coefficients and recalibration coefficients. The ultimate aim was to derive the bias from January 2018 to March 2025. As demonstrated in Fig. 9, recalibration leads to a substantial enhancement in the accuracy of calibration in the VIS and NIR bands. When combined with the inherent calibration error of MODIS, the recalibrated accuracy is controlled within a range of 1%–2%. The average across seven channels is close to 1%.

As illustrated in Fig. 9, the discrepancy between the seven calibration sets was slight in the initial stages, particularly B05 (1.38 μm). However, by the conclusion of the designated operational lifespan, specifically after 2022, the disparities in calibration uncertainties became substantially less pronounced in comparison to those observed in operational calibration. Subsequent calculations of the reflectance differences demonstrate that the overall accuracy variation across the seven RSBs of MERSI-II remains relatively stable. Initially, the discrepancy is confined within a range of 1%–1.5%, but it expands to approximately 3% after 2021 or 2022. This finding indicates that the recalibration process implemented in this study has a substantial impact on enhancing the quality of satellite data, resulting in an overall accuracy enhancement of approximately 2%.

As is well known, the primary applications of RSBs data are focused on cloud detection and vegetation detection [44]. To evaluate the application-level reliability of the recalibration coefficients for FY-3D/MERSI-II, vegetation indices derived from calibrated reflectance products are employed as independent validation metrics. In this study, both the normalized difference vegetation index (NDVI) and the enhanced vegetation index 2 (EVI2) are used. NDVI remains the most widely adopted vegetation index in operational products and long-term climate applications, while EVI2 is less prone to saturation under high-biomass conditions and provides enhanced sensitivity over densely vegetated surfaces.

Following extensive sample screening, it was found that over regions with minimal vegetation coverage, such as deserts and ice or snow fields, NDVI values derived from the operational and recalibrated coefficients are highly consistent, and both calibration schemes are capable of reliably identifying these highly reflective surfaces. To better assess calibration impacts under conditions most sensitive to NIR radiometric fidelity, the present study therefore focuses on densely vegetated regions, including tropical rainforests and subtropical monsoon forest ecosystems. The final validation regions selected are Southeast Asia and the Guangxi–Guangdong region of southern China.

For the regions previously specified, the NDVI values should be calculated. The identification of areas with the greatest discrepancies is achieved by performing recalibration and comparing the observed values with the recalibrated

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:17:06 UTC from IEEE Xplore. Restrictions apply.

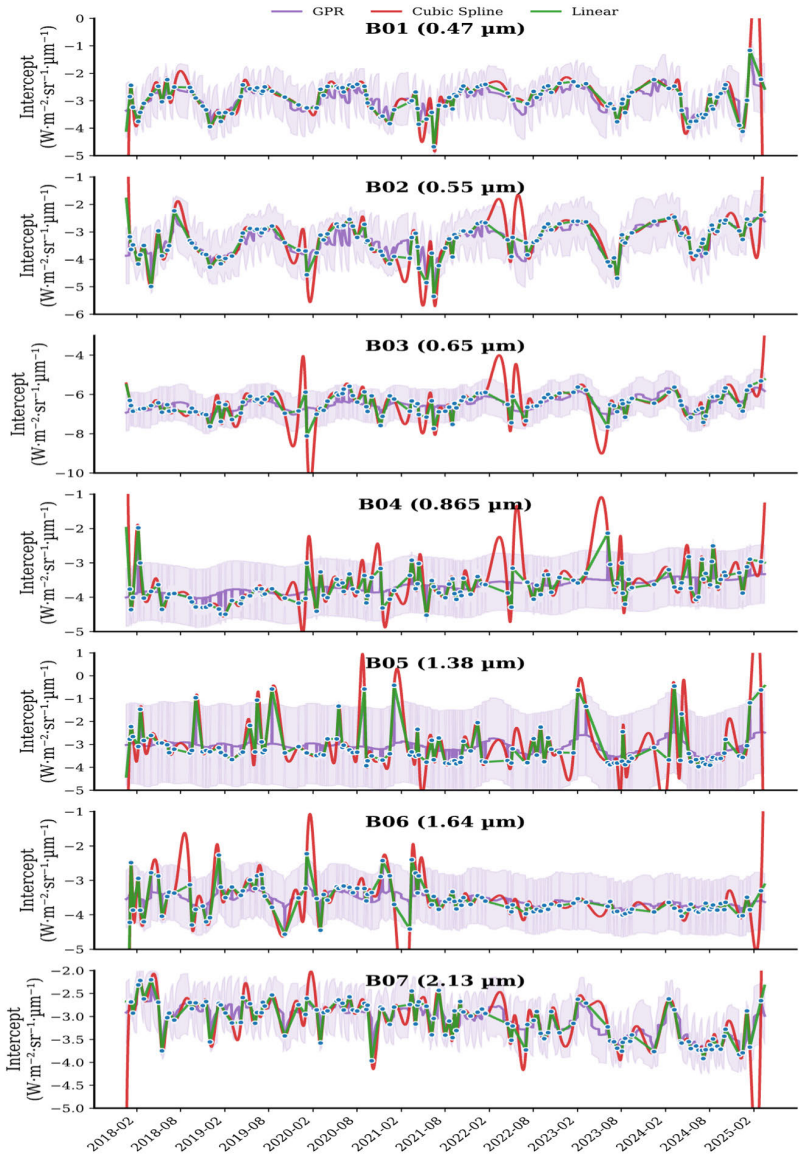


Fig. 6. Comparison of three interpolation methods. Blue dots denote recalibrated observations. The purple curve shows GPR estimates, with the light-purple band indicating the 95% confidence interval. The red curve shows cubic-spline interpolation, and the green curve shows linear interpolation.

values. The calculation of the NDVI is as follows:

$$NDVI = \frac{\rho_{NIR} - \rho_{RED}}{\rho_{NIR} + \rho_{RED}} \tag{13}$$

where ρ_{NIR} is the reflectance in the NIR B04 (0.86 μm), and ρ_{RED} is the reflectance in the Red B03 (0.65 μm).

For each index, the upper panel displays the spatial distribution of index differences over the full study area, and the lower panels show two sub-regions with the largest absolute differences, both located within Southeast Asia.

To mitigate the well-known saturation limitation of NDVI and provide a more robust assessment over densely vegetated areas, EVI2 is additionally calculated as

$$EVI2 = 2.5 \times \frac{\rho_{NIR} - \rho_{RED}}{\rho_{NIR} + 2.4\rho_{RED} + 1} \tag{14}$$

which reduces saturation effects and suppresses background and atmospheric influences, thereby offering improved sensitivity under high vegetation cover.

Fig. 10 illustrates the impact of recalibration on the application-level performance of FY-3D/MERSI-II vegetation indices on June 6, 2023. The main panel shows the spatial differences between vegetation indices derived using the operational and recalibrated coefficients, while the lower subregion highlights Southeast Asia. Large NDVI differences are primarily observed over nonvegetated or sparsely vegetated surfaces with low baseline NDVI values. These differences do not indicate actual surface changes but reflect the inherent mathematical sensitivity of NDVI under low-signal conditions, where small reflectance perturbations can be significantly amplified.

Over densely vegetated regions, where NDVI is known to approach saturation, vegetation indices derived using the recalibrated coefficients exhibit improved sensitivity to biomass gradients and better resolve spatial variability. This improvement becomes particularly evident during the later years of the FY-3D mission, when sensor response drift would

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:17:06 UTC from IEEE Xplore. Restrictions apply.

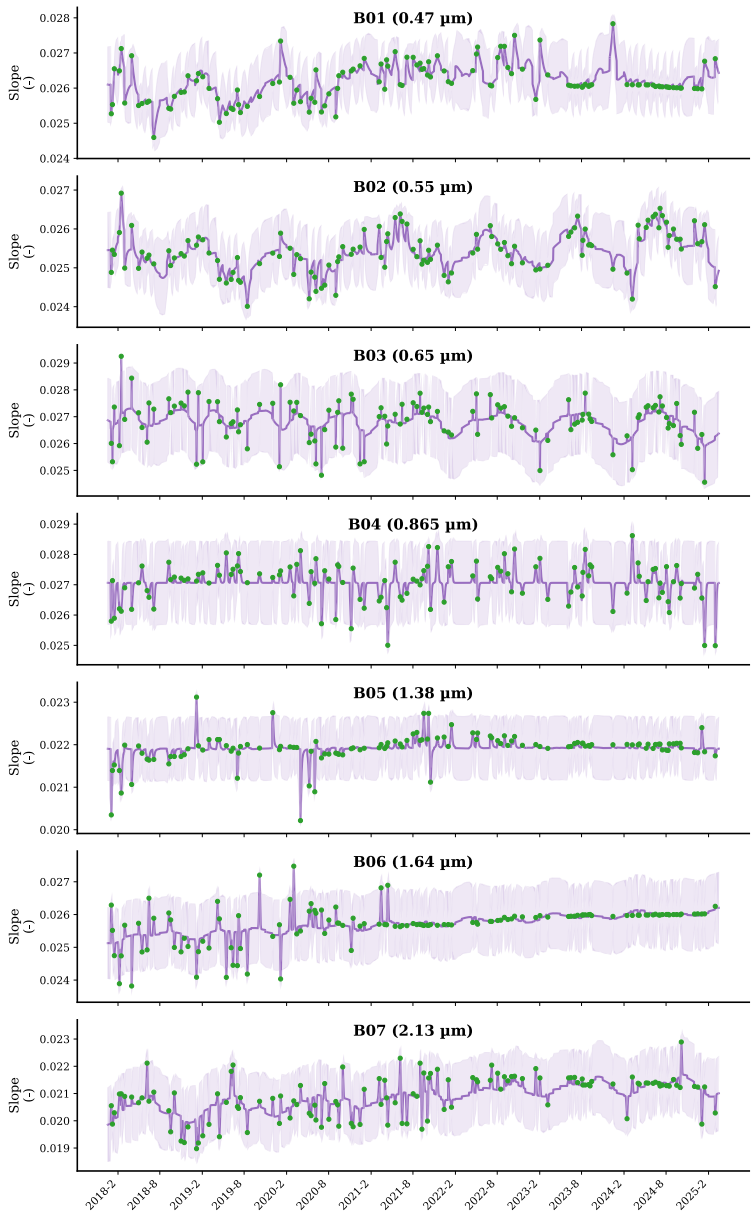


Fig. 7. Interpolated time series of recalibration slopes for FY-3D/MERSI-II Bands 1–7. Green points indicate the raw data, the solid purple curve is the interpolation results of slope, and the lavender shading denotes the confidence interval.

otherwise suppress vegetation contrast in NDVI products derived from the operational coefficients. Importantly, consistent behavior is observed in the EVI2-based results, which demonstrate reduced saturation and more stable responses over high-biomass regions. The agreement between NDVI- and EVI2-based analyses confirms that the observed improvements are not artifacts of NDVI saturation, but instead reflect a genuine enhancement in NIR radiometric performance achieved through recalibration.

V. Discussion

A. Impact of Aqua/Modis Orbital Drift on Recalibration Robustness

Since December 2021, Aqua has operated in free-drift mode, leading to progressive changes in solar illumination geometry and atmospheric path length relative to FY-3D/MERSI-II overpasses. To mitigate potential time-dependent biases, strict spatiotemporal, angular, and atmospheric constraints were applied during SNO pairing. Under these controls, the recalibration coefficients exhibit smooth temporal evolution rather than fluctuations synchronized with Aqua's orbital drift.

To further assess the potential influence of the free-drift orbit, the matchup dataset was conceptually examined by distinguishing periods before and after the onset of Aqua's orbital drift. The analysis indicates that the recalibration slopes remain highly stable across the two periods, with variations well within the expected statistical uncertainty. This behavior suggests that the radiometric gain relationship between MERSI-II and MODIS is largely insensitive to Aqua's orbital drift.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:17:06 UTC from IEEE Xplore. Restrictions apply.

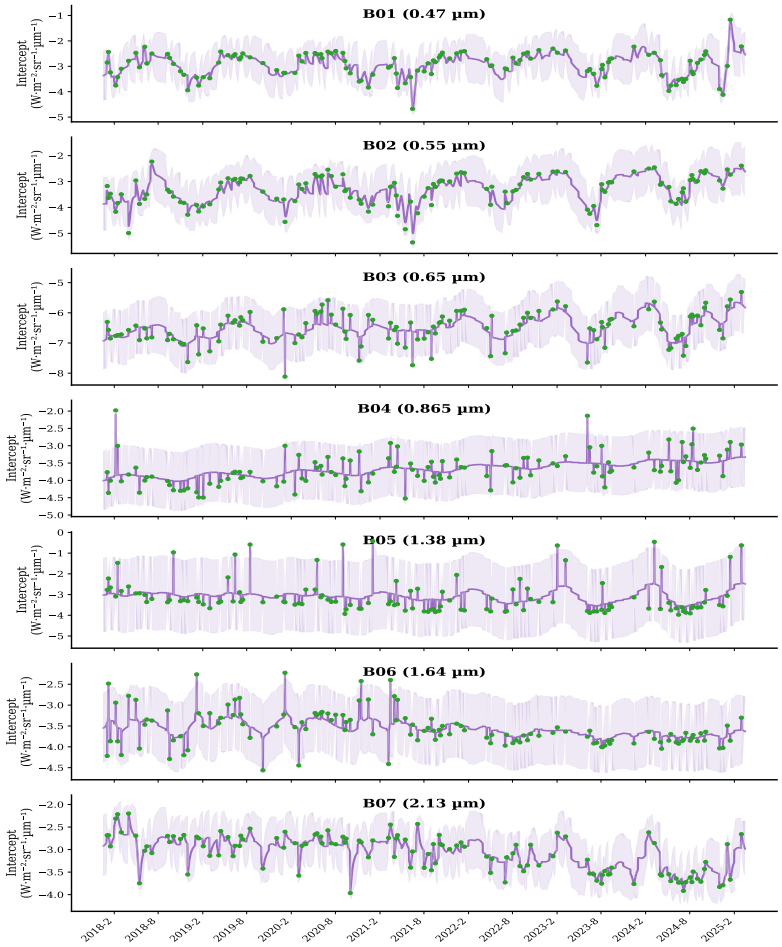


Fig. 8. Same as Fig. 7, but for the intercept.

In contrast, relatively larger variations are observed in the intercept terms. This behavior is physically consistent with the effect of changing solar illumination geometry, which primarily introduces additive atmospheric path radiance and background contributions. Such additive components are preferentially absorbed by the intercept term, while the multiplicative gain (slope) remains largely unaffected.

Notably, abrupt changes in calibration performance are observed in multiple MERSI-II bands around early 2020, nearly two years prior to Aqua's transition to free-drift mode. This temporal mismatch further indicates that the dominant driver of bias evolution is intrinsic MERSI-II instrument aging, rather than reference sensor orbital perturbations.

Together, these results demonstrate that Aqua's orbital drift exerts a secondary influence on the recalibration outcome, while the long-term bias evolution is governed primarily by target sensor degradation processes.

B. Residual Biases and Seasonal Variability

In spite of the substantial bias reduction achieved through recalibration, residual seasonal variations persist in several bands, particularly B01, B04, and B06 (Fig. 9). These seasonal signatures are consistent with the fluctuations observed in the radiometric bias time series shown in Fig. 5, suggesting a common underlying mechanism rather than independent artifacts.

Importantly, these seasonal variations cannot be fully explained by atmospheric absorption effects. Under the strict SNO pairing criteria applied in this study, atmospheric path length and thermodynamic conditions are tightly constrained, and the SWIR bands (B05–B07), which are intrinsically more sensitive to atmospheric absorption, exhibit comparatively weaker seasonal fluctuations. This behavior indicates that the dominant driver of the observed VIS/NIR seasonality is not atmospheric variability. It is worth noting that similar seasonal fluctuations and residual radiometric variability have been widely reported for FY-3 series sensors. Xu et al. [27] demonstrated that FY-3A/B/C VIS and NIR channels exhibit persistent seasonal and long-term variability even after operational calibration and vicarious correction. This suggests that the residual behavior observed in B01, B04, and B06 reflects intrinsic sensor performance characteristics, rather than limitations of the present recalibration framework.

Instead, the seasonal fluctuations in the VIS and NIR bands are primarily attributed to viewing geometry modulation and surface anisotropy effects. Although angular constraints are applied during collocation, the relative VZA between MERSI-II and Aqua MODIS exhibits a clear seasonal pattern

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:17:06 UTC from IEEE Xplore. Restrictions apply.

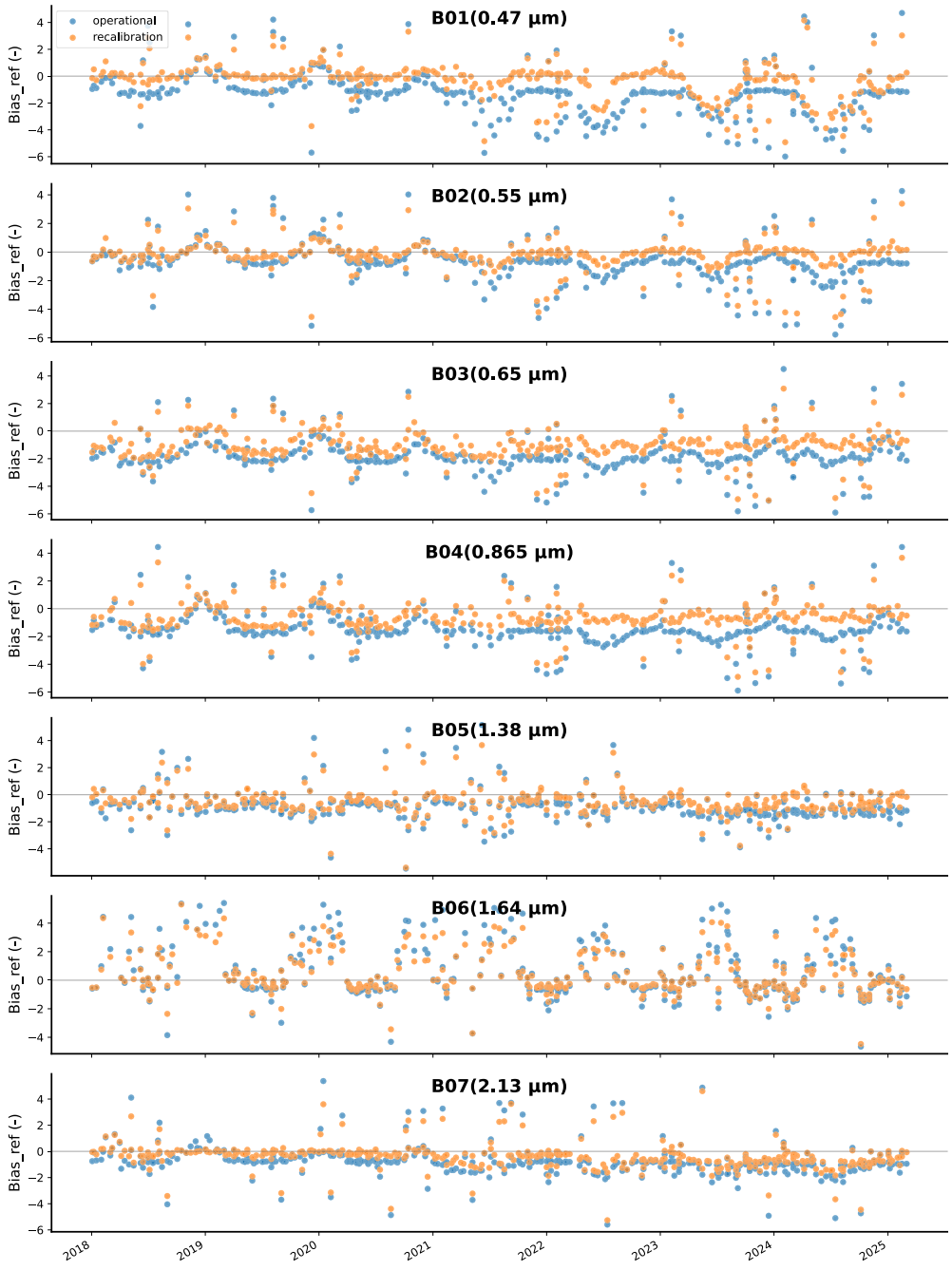


Fig. 9. Time series distributions of FY-3D/MERSI-II B01–B07 calibration bias [(MERSI-II reflectance–MODIS reflectance)/MODIS reflectance ×100%], where orange points represent recalibration results and blue points represent operational calibration results.

due to orbital geometry and latitude-dependent sampling. Supplementary analysis of the intersensor VZA differences (Supplementary Figs. S4 and S5) shows that the monthly distribution and range of VZA vary systematically throughout the year.

For VIS/NIR bands such as B01 and B04, surface reflectance is highly anisotropic and strongly influenced by vegetation structure and phenological state. Seasonal changes in land surface properties, combined with varying illumination–viewing geometry, lead to residual BRDF-related biases that are not fully corrected by linear radiometric recalibration. In contrast, SWIR bands are less sensitive to surface BRDF effects, resulting in comparatively weaker seasonal modulation.

For B06 (1.64 μm), residual biases are additionally influenced by incomplete SRF correction near water vapor absorption features, where minor spectral mismatches can be amplified under varying atmospheric humidity.

Potential mitigation strategies include incorporating BRDF-normalized reflectance products, enhancing angular stratification during regression, and adopting seasonally adaptive correction schemes. These approaches are expected to further suppress geometry-induced residual biases in future implementations.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:17:06 UTC from IEEE Xplore. Restrictions apply.

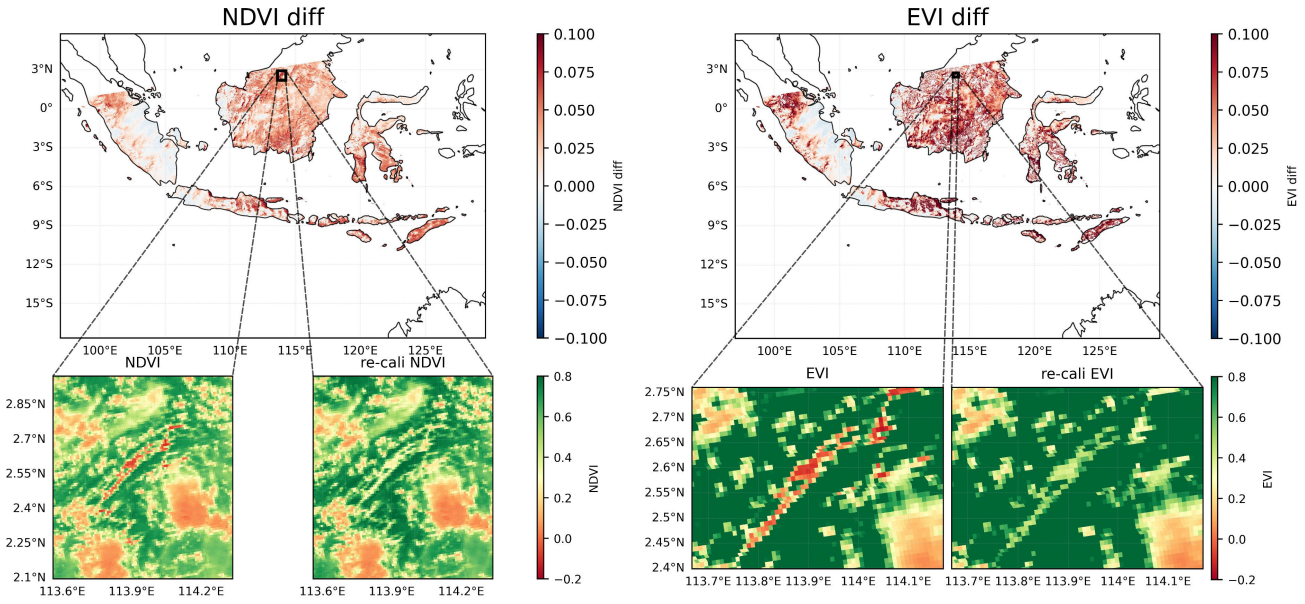


Fig. 10. Application-level comparison of FY-3D/MERSI-II vegetation indices derived using two sets of calibration coefficients (operational versus recalibrated) on June 6, 2023. The left column shows the differences in NDVI, while the right column presents the corresponding differences in EVI2.

C. Long-Term Degradation Behavior of FY-3D/MERSI-II

A pronounced increase in calibration bias after 2021–2022 coincides with FY-3D/MERSI-II operating beyond its nominal five-year design life, suggesting that the observed bias evolution primarily reflects intrinsic instrument aging. Earlier degradation in B05 (1.38 μm) and B06 (1.64 μm) around October 2021 is attributed to optical contamination and the heightened sensitivity of these bands to minor spectral or optical changes. For other bands, abrupt variations observed after May 2024 are consistent with environmental noise fluctuations and cumulative component fatigue, including gradual detector responsivity loss and long-term thermal control drift. A definitive diagnosis of the individual contributions from these mechanisms would require detailed on-orbit telemetry and instrument health data, which are beyond the scope of the present study.

In contrast, the Terra and Aqua MODIS instruments have demonstrated exceptional radiometric stability over more than two decades of operation, largely attributed to conservative optical and thermal design and robust onboard calibration systems. While a direct engineering comparison is beyond the scope of this study, the contrast highlights the importance of lifetime-oriented instrument design and calibration planning for future generations of moderate-resolution imagers.

D. Implications for Calibration Strategy and Data Continuity

The post-2021 aging behavior of MERSI-II underscores the necessity of adaptive, high-frequency recalibration strategies during extended mission phases. Compared with standard operational calibration updates, which are typically performed at multimonth intervals, frequent recalibration more effectively captures short-term radiometric variations and mitigates the impact of satellite aging and orbital drift.

The SFP model proposed in this study corrects radiometric bias (mean relative bias $\approx$ 1%) while preserving the native SRFs of MERSI-II, a critical advantage for water-vapor-sensitive bands. Application-level validation based on NDVI demonstrates improved consistency and sensitivity to biotic gradients. The resulting seven-year recalibration record provides valuable insight into long-term sensor performance beyond nominal mission life and supports the development of degradation-aware calibration strategies for future Fengyun satellite missions.

Looking forward, future generations of Chinese moderate-resolution imagers may benefit from advances in detector technology. In particular, InGaAs detectors have demonstrated superior long-term radiometric stability compared to HgCdTe detectors in the NIR spectral region, with reduced susceptibility to responsivity drift and thermal sensitivity. The adoption of such detector technologies, together with enhanced onboard calibration and thermal control strategies, has the potential to further improve radiometric stability and extend effective mission lifetime.

VI. Conclusion

In this study, we develop a recalibration algorithm for the FY-3D/MERSI-II long-term record (January 2018–March 2025). Extensive MODTRAN forward simulations are used to derive a new SFP model that links FY-3D/MERSI-II (target) to the AQUA/MODIS reference while preserving the target sensor's spectral observation characteristics. The SFP model is embedded in the algorithm, and rigorously screened high-frequency SNO events between the two platforms are used for cross calibration. The resulting algorithm substantially improves the radiometric calibration of the RSBs and, with a high-confidence interpolation scheme, yields a stable and reliable daily calibration record. The primary conclusions of this study are summarized as follows.

1) MK test analysis of the recalibrated coefficient time series shows that all seven RSBs of FY-3D/MERSI-II

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:17:06 UTC from IEEE Xplore. Restrictions apply.

have significant instrument attenuation breakpoints, further confirming the overall radiometric response degradation of the instrument. Among them, B01 (0.47 μm), B02 (0.55 μm) and B04 (0.86 μm) exhibit obvious positive trends, indicating a continuous decline in their radiometric response, and are accompanied by environmental noise fluctuations after May 2024 without a single clear mutation time; B03 (0.65 μm) shows a negative trend, manifesting as abnormal changes in radiometric response, and its mutation time is synchronized with the environmental noise fluctuations after May 2024.

2) A high-frequency recalibration framework using GPR enables daily updates of calibration coefficients (surpassing the current state-of-the-art of monthly or semi-annual updates). Validation shows GPR interpolation reduces errors by ~40% compared with linear interpolation, ensuring reliable daily coefficient generation that captures short-term radiometric degradation of MERSI-II.
3) After recalibration using AQUA/MODIS as reference, the radiometric calibration errors of seven RSBs of FY-3D/MERSI-II were controlled within 1%–2%. Compared with the 3%–5% calibration error range of operational calibration coefficients, the overall accuracy across the entire period of FY-3D/MERSI-II RSBs was improved by ~2% (Fig. 9); particularly after 2022 (the satellite's postdesign lifespan phase), the accuracy of B01 (0.47 μm) and B04 (0.86 μm) increased by about 3%.

The long-term recalibrated RSB data from FY-3D/MERSI-II provide substantial value for ecological and atmospheric remote sensing, supporting applications such as cloud detection and vegetation dynamics monitoring. Looking ahead, we will pursue high-frequency recalibration across the Fengyun-3 series satellite, and we will also extend the new SFP model toward quantitative applications involving additional satellite sensors and datasets. The new long-term recalibration coefficients (including slope and intercept) for FY-3D/MERSI-II seven RSBs from January 2018 to March 2025, along with the GPR interpolation methods described in this article, are available at https://github.com/Rhett12345/mersi_recalibration. The repository will be updated on a monthly or quarterly basis.

Data Availability Statement

The calibration coefficients are updated monthly or quarterly, and the corresponding daily interpolation results are publicly available in two data formats, CSV and HDF. Detailed information regarding the recalibration coefficients of the FY-3D/MERSI-II instrument, as well as calibration coefficients for other satellite sensors, can be accessed through the following GitHub repository: https://github.com/Rhett12345/mersi_recalibration.

The FY-3D MERSI-II GEO and L1B datasets used in this study are available from the NSMC data portal at https://satellite.nsmc.org.cn/DataPortal/cn/home/index.html.

The GEO and L1B datasets of the AQUA MODIS sensor can be obtained from NASA's Earthdata Search system at https://search.earthdata.nasa.gov/search/granules?p = C1379759127-LAADS&pg [0][v] = f&pg [0][gsk] = -start_date&q = MYD02.

Orbital satellite data for simulating orbits and calculating co-visibility observations can be requested on this website: https://celestrak.org/satcat/search.php.

Acknowledgment

The authors' gratitude is extended to the National Aeronautics and Space Administration (NASA) for providing openly accessible and readily usable MODIS Level-1B data for the recalibration work. Additionally, we extend our profound gratitude to the high-performance computing (HPC) cluster at the School of Atmospheric Sciences, Sun Yat-sen University, for its provision of high-quality and reliable computational resources during the execution of radiative transfer simulations and subsequent calibrations.

References

[1] K. Berger et al., "Multi-sensor spectral synergies for crop stress detection and monitoring in the optical domain: A review," *Remote Sens. Environ.*, vol. 280, Oct. 2022, Art. no. 113198, doi: 10.1016/j.rse.2022.113198.
[2] B. Dechant et al., "NIRVP: A robust structural proxy for sun-induced chlorophyll fluorescence and photosynthesis across scales," *Remote Sens. Environ.*, vol. 268, Jan. 2022, Art. no. 112763, doi: 10.1016/j.rse.2021.112763.
[3] H. Letu et al., "A new benchmark for surface radiation products over the East Asia–Pacific region retrieved from the himawari-8/AHI next-generation geostationary satellite," *Bull. Amer. Meteorological Soc.*, vol. 103, no. 3, pp. E873–E888, Mar. 2022, doi: 10.1175/bams-d-20-0148.1.
[4] P. Xia et al., "Accurate nowcasting of cloud cover at solar photovoltaic plants using geostationary satellite images," *Nature Commun.*, vol. 15, no. 1, p.510, 2024, doi: 10.1038/s41467-023-44666-1.
[5] D. Xian, P. Zhang, L. Gao, R. Sun, H. Zhang, and X. Jia, "Fengyun Meteorological Satellite products for Earth system science applications," *Adv. Atmos. Sci.*, vol. 38, no. 8, pp. 1267–1284, Aug. 2021, doi: 10.1007/s00376-021-0425-3.
[6] X. Wu, T. Hewison, and Y. Tahara, "GSICS GEO-LEO intercalibration: Baseline algorithm and early results," *Proc. SPIE*, vol. 745604, pp. 25–36, Aug. 2009, doi: 10.1117/12.825460.
[7] X. Wu, "Operational calibration of solar reflectance channels of the Advanced Very High Resolution Radiometer (AVHRR)," *Proc. SPIE*, vol. 5542, pp. 272–280, Oct. 2004, doi: 10.1117/12.561289.
[8] N. Xu et al., "Prelaunch calibration and radiometric performance of the advanced MERSI II on FengYun-3D," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4866–4875, Aug. 2018, doi: 10.1109/TGRS.2018.2841827.
[9] X. Xiong et al., "Results and lessons from a decade of Terra MODIS on-orbit spectral characterization," *Proc. SPIE*, vol. 78620, pp. 159–169, Oct. 2010, doi: 10.1117/12.868930.
[10] X. Xiong et al., "Aqua MODIS: 20 years of on-orbit calibration and performance," *J. Appl. Remote Sens.*, vol. 17, no. 3, Aug. 2023, Art. no. 037501, doi: 10.1117/1.jrs.17.037501.
[11] X. Xiong, V. Salomonson, W. Barnes, B. Guenther, X. Xie, and J. Sun, "An overview of Terra MODIS reflective solar bands on-orbit calibration," in *Proc. IEEE Int. Symp. Geosci. Remote Sens.*, Denver, CO, USA, Jul. 2006, pp. 1111–1114, doi: 10.1109/IGARSS.2006.287.
[12] X. Xiong et al., "MODIS reflective solar bands on-orbit calibration and performance," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6355–6371, Sep. 2019, doi: 10.1109/TGRS.2019.2905792.
[13] T. Choi, C. Cao, S. Blonski, W. Wang, S. Uprety, and X. Shao, "NOAA-20 VIIRS reflective solar band postlaunch calibration updates two years in-orbit," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7633–7642, Nov. 2020, doi: 10.1109/TGRS.2020.2982764.
[14] H. Oudrari et al., "JPSS-1 VIIRS radiometric characterization and calibration based on pre-launch testing," *Remote Sens.*, vol. 8, no. 1, p. 41, Jan. 2016, doi: 10.3390/rs8010041.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:17:06 UTC from IEEE Xplore. Restrictions apply.

[15] X. Xiong et al., "VIIRS on-orbit calibration methodology and performance," *J. Geophys. Res., Atmos.*, vol. 119, no. 9, pp. 5065–5078, May 2014, doi: 10.1002/2013jd020423.
[16] K. Twedt et al., "On-orbit calibration and performance of NOAA-20 VIIRS reflective solar bands," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 1001413, doi: 10.1109/TGRS.2021.3108970.
[17] J. V. Servera et al., "Systematic assessment of MODTRAN emulators for atmospheric correction," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4101917, doi: 10.1109/TGRS.2021.3071376.
[18] R. Bhatt, D. R. Doelling, D. Morstad, B. R. Scarino, and A. Gopalan, "Desert-based absolute calibration of successive geostationary visible sensors using a daily exoatmospheric radiance model," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 6, pp. 3670–3682, Jun. 2014, doi: 10.1109/TGRS.2013.2274594.
[19] P. Zhang et al., "Development of the Chinese space-based radiometric benchmark mission LIBRA," *Remote Sens.*, vol. 12, no. 14, p. 2179, Jul. 2020, doi: 10.3390/rs12142179.
[20] Y. Gong, H. Li, X. Hu, and H. Shen, "Cross validation of FY3D MWRI passive microwave LST with MODIS LST under clear-sky conditions," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 10786–10802, 2025, doi: 10.1109/JSTARS.2025.3560132.
[21] L. Chen, X. Hu, N. Xu, and P. Zhang, "The application of deep convective clouds in the calibration and response monitoring of the reflective solar bands of FY-3A/MERSI (Medium resolution spectral Imager)," *Remote Sens.*, vol. 5, no. 12, pp. 6958–6975, Dec. 2013, doi: 10.3390/rs5126958.
[22] L. Sun, X. Hu, N. Xu, J. Liu, L. Zhang, and Z. Rong, "Postlaunch calibration of FengYun-3B MERSI reflective solar bands," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1383–1392, Mar. 2013, doi: 10.1109/TGRS.2012.2217345.
[23] J. R. Eyre et al., "Assimilation of satellite data in numerical weather prediction. Part II: Recent years," *Quart. J. Roy. Meteorological Soc.*, vol. 148, no. 743, pp. 521–556, Jan. 2022, doi: 10.1002/qj.4228.
[24] F. Zhang, W. Chen, X. Hu, X. Zhang, H. Tang, and B. Peng, "On-orbit degradation tracking of FY-3D/MERSI-II based on ocean glint," *IEEE Sensors J.*, vol. 24, no. 12, pp. 19175–19189, Jun. 2024, doi: 10.1109/JSEN.2024.3391287.
[25] L. Chen, N. Xu, Y. Li, R. Wu, X. Hu, and P. Zhang, "FY-3C/MERSI prelaunch calibration for reflective solar bands," *Proc. SPIE*, vol. 92640, pp. 221–229, Nov. 2014, doi: 10.1117/12.2070397.
[26] L. Sun, X. Hu, M. Guo, and N. Xu, "Multisite calibration tracking for FY-3A MERSI solar bands," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 12, pp. 4929–4942, Dec. 2012, doi: 10.1109/TGRS.2012.2215613.
[27] N. Xu et al., "Recalibration and reprocessing of the long-term FY-3 MERSI historical data," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5640715, doi: 10.1109/TGRS.2024.3432899.
[28] Z. Wang, X. Xiong, T. Choi, and D. Link, "On-orbit characterization of MODIS modulation transfer function using the moon," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 7, pp. 4112–4121, Jul. 2014, doi: 10.1109/TGRS.2013.2279610.
[29] F. Wang, C. Liu, B. Yao, X. Hu, P. Zhang, and B.-J. Sohn, "Cloud-target calibration for fengyun-3D MERSI-II solar reflectance bands: Model development and instrument stability," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5401313, doi: 10.1109/TGRS.2023.3244949.
[30] T. J. Hewison et al., "GSICS inter-calibration of infrared channels of geostationary imagers using Metop/IASI," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1160–1170, Mar. 2013, doi: 10.1109/TGRS.2013.2238544.
[31] K. Wang and R. E. Dickinson, "Global atmospheric downward longwave radiation at the surface from ground-based observations, satellite retrievals, and reanalyses," *Rev. Geophysics*, vol. 51, no. 2, pp. 150–185, Apr. 2013, doi: 10.1002/rog.20009.
[32] Y. Chen, K. Sun, W. Li, X. Hu, P. Li, and T. Bai, "Vicarious calibration of FengYun-3D MERSI-II at Railroad Valley Playa site: A case for sensors with large view angles," *Remote Sens.*, vol. 13, no. 7, p. 1347, Apr. 2021, doi: 10.3390/rs13071347.
[33] J. Gorroño, A. C. Banks, N. P. Fox, and C. Underwood, "Radiometric inter-sensor cross-calibration uncertainty using a traceable high accuracy reference hyperspectral imager," *ISPRS J. Photogramm. Remote Sens.*, vol. 130, pp. 393–417, Aug. 2017, doi: 10.1016/j.isprsjprs.2017.07.002.
[34] A. Nghien, M. Lei, M. Brédif, and O. Hagolle, "Radiometric cross-calibration of an aerial sensor with satellite top-of-atmosphere reflectance," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. XLVIII-M-7-2025, pp. 49–55, May 2025, doi: 10.5194/isprs-archives-xlviii-m-7-2025-49-2025.
[35] P. Zhang, K. Holmlund, M. Goldberg, and J. Lafeuille, "The global space-based inter-calibration system (GSICS)," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Beijing, China, Jul. 2016, pp. 5522–5523, doi: 10.1109/IGARSS.2016.7730440.
[36] L. R. Porto, N. N. Imai, A. M. G. Tommaselli, and L. S. Neto, "Radiometric calibration performance of a multispectral camera with a single sensor and multiple heads," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 43, pp. 17–22, May 2022, doi: 10.5194/isprs-archives-xliii-b1-2022-17-2022.
[37] J. Vicent et al., "Comparative analysis of atmospheric radiative transfer models using the atmospheric look-up table generator (ALG) toolbox (version 2.0)," *Geoscientific Model Develop.*, vol. 13, no. 4, pp. 1945–1957, Apr. 2020, doi: 10.5194/gmd-13-1945-2020.
[38] A. M. Sayer, N. C. Hsu, and C. Bettenhausen, "Implications of MODIS bow-tie distortion on aerosol optical depth retrievals, and techniques for mitigation," *Atmos. Meas. Techn.*, vol. 8, no. 12, pp. 5277–5288, Dec. 2015, doi: 10.5194/amt-8-5277-2015.
[39] M. Min, B. Chen, N. Xu, X. He, X. Wei, and M. Wang, "Nonnegligible diurnal and long-term variation characteristics of the calibration biases in fengyun-4A/AGRI infrared channels based on the oceanic drifter data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5408415, doi: 10.1109/TGRS.2022.3160450.
[40] I. E. Gordon et al., "The HITRAN2020 molecular spectroscopic database," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 277, Jan. 2022, Art. no. 107949, doi: 10.1016/j.jqsrt.2021.107949.
[41] S. Vasilchenko, H. Tran, D. Mondelain, S. Kassi, and A. Campargue, "Accurate absorption spectroscopy of water vapor near 1.64 μm in support of the MEthane remote LiDAR missioN (MERLIN)," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 235, pp. 332–342, Sep. 2019, doi: 10.1016/j.jqsrt.2019.06.027.
[42] K. H. Hamed, "Trend detection in hydrologic data: The mann–Kendall trend test under the scaling hypothesis," *J. Hydrol.*, vol. 349, nos. 3–4, pp. 350–363, Feb. 2008, doi: 10.1016/j.jhydrol.2007.11.009.
[43] C. Lyu, X. Liu, and L. Mihaylova, "Review of recent advances in Gaussian process regression methods," 2024, *arXiv:2409.08112*.
[44] J. Liu et al., "Near-real-time atmospheric and oceanic science products of himawari-8 and himawari-9 geostationary satellites over the South China Sea," *Earth Syst. Sci. Data*, vol. 16, no. 10, pp. 4949–4969, Oct. 2024, doi: 10.5194/essd-16-4949-2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:17:06 UTC from IEEE Xplore. Restrictions apply.